

Gursimran Singh

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RESEARCH INTERESTS

In-network computation and data-plane programming (P4); programmable data planes for network security; network and Internet measurement; network security and censorship measurement.

EDUCATION

Oklahoma State University — Ph.D. in Computer Science Expected 2028
Advisor: Dr. H. B. Acharya (hbhatta@okstate.edu).

Rochester Institute of Technology — Ph.D. coursework, Computer and Information Science (GPA 3.80/4.0) Transferred

State University of New York (SUNY) at Buffalo — M.S., Computer and Information Science (GPA 3.52/4.0) 2021

PUBLICATIONS & SUBMISSIONS

Under review, ICNP 2026 — work on programmable switches and network security.
Programmable Data Planes for Network Security — position paper systematizing recent advances in security applications built on programmable switches. NetCoM, July 2025.
POSTER: A Cyberspace Study of the Russia-Ukraine War — measurement findings on the conflict's impact on Internet resilience. ACM ASIACCS, July 2023.

ACADEMIC & PROFESSIONAL SERVICE

- **Technical Program Committee member**, ACM S3 2026 Workshop (invited and accepted).
- **President**, Computer Science Graduate Student Association (CSGSA), Oklahoma State University — current.
- **Council Member**, Doctoral Student Association (DSA), Rochester Institute of Technology — organized weekly events and career workshops for Ph.D. students (2025).
- **Vice-President**, CSEGSU, University at Buffalo — organized workshops and competitions, including a Conversational AI special-interest group (2020–2021).
- **Mentorship** — mentored 3 master's and 2 undergraduate students on research and capstone projects (2022–2025).

RESEARCH EXPERIENCE

In-Network Computation Using Data-Plane Programming in P4 2021–Present

- Developing high-performance packet processors comparable to application-layer firewalls and network intrusion detection systems.
- Implementing advanced algorithms for packet analysis and filtering in P4 on the Intel Tofino (TNA) platform.

Internet Censorship Measurement 2021–Present

- Designed and implemented tools for detecting and measuring Internet censorship across multiple countries.
- Published measurement findings on the Russia-Ukraine conflict's impact on Internet resilience (ASIACCS 2023).

TEACHING EXPERIENCE

Adjunct Instructor, RIT — CSEC-140: Introduction to Cybersecurity 2022, 2025

- Delivered lectures and hands-on virtual labs covering authentication, data security, risk management, network and system security, application security, and human factors.

Adjunct Instructor, RIT — CSCI-455: Principles of Cybersecurity 2024, 2025

- Instructed on cybersecurity principles, system protection, risk management, and ethical considerations for building secure, trusted systems.

Adjunct Instructor, RIT — GCIS-123: Software Development & Problem Solving I 2025

- Taught computational problem solving in Python: searching and sorting, file I/O, lists, classes, and dictionaries.

Teaching Assistant, RIT — CSCI-243 (Mechanics of Programming); CSEC-759 (Grad. Seminar on Advanced Networking & Distributed Systems Security) 2023–2024

Teaching Assistant, Oklahoma State University — Distributed and Cloud Computing Spring 2026

SELECTED PROJECTS

- **Networking components at application & transport layers** — implemented reliable transport protocols (Alternating-Bit, Go-Back-N, Selective-Repeat) and a multi-client TCP chat system. [C/C++]
- **Task offloading for QoS optimization with a dynamic ANN** — prototype that adapts at run time to query parameters (max-hops, query volume) to optimize QoS. [C, C++, CUDA]
- **Core OS components in Pintos** — kernel-level threading and system-call interfaces. [C/Pintos]

TECHNICAL SKILLS

Languages: C, C++, P4, Python, Java, Shell scripting.

Networking & Security: Tofino Native Architecture (TNA), Scapy, Wireshark, Zmap, Nmap, Aircrack-ng, Kali Linux, OllyDbg, IDA.

ML & Data: Scikit-learn, Keras, NumPy, Pandas.

CERTIFICATIONS

- KLCP (2023) and OSWP (2023), Offensive Security.
- Intel Tofino / P4: ICA-XFG-101 (Intro to P416, Tofino Family, P4 Studio SDE, 2023); ICA-XFG-201–206 (Advanced TNA, incl. parsing/deparsing, recirculation, mirroring, 2024).
- Google Cloud: Introduction to Generative AI, Responsible AI, and LLMs (2023).